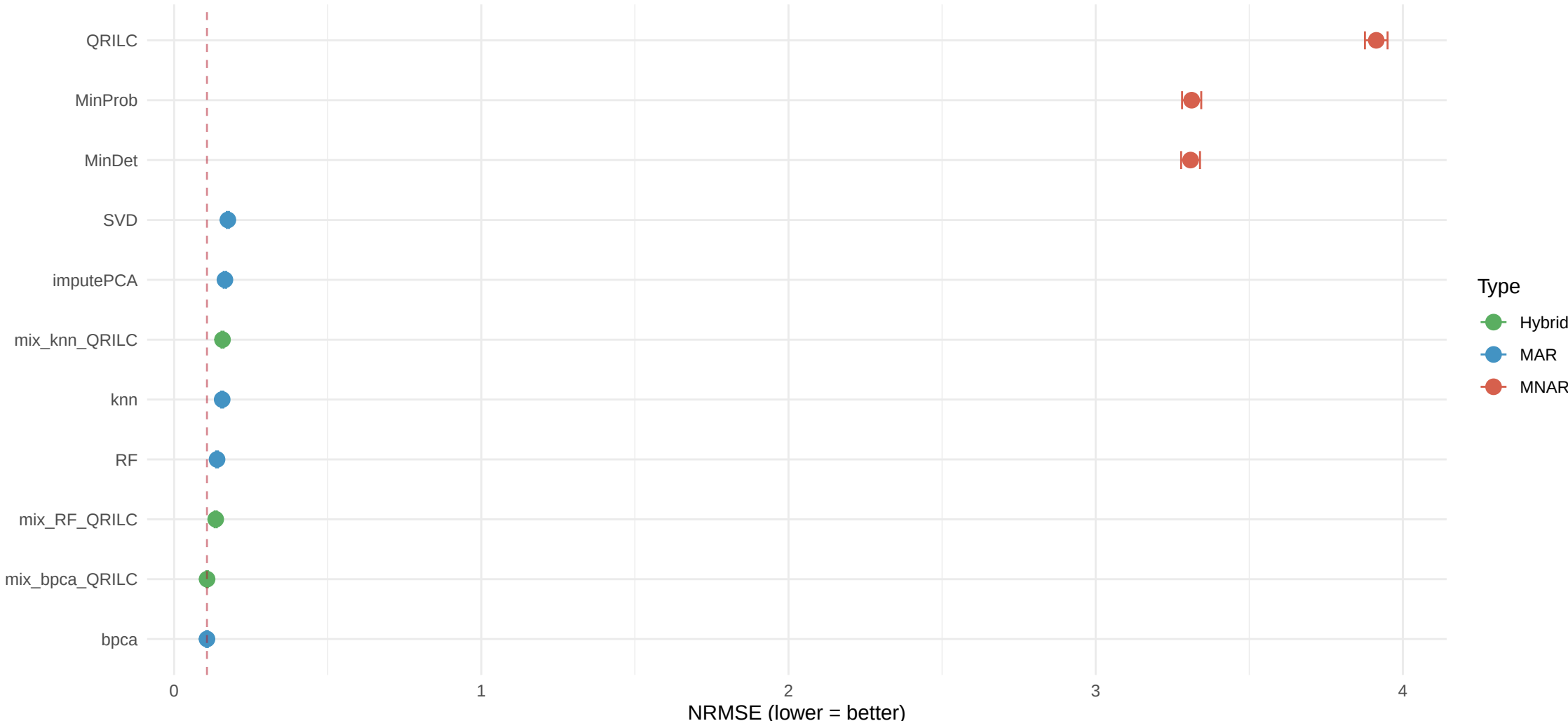


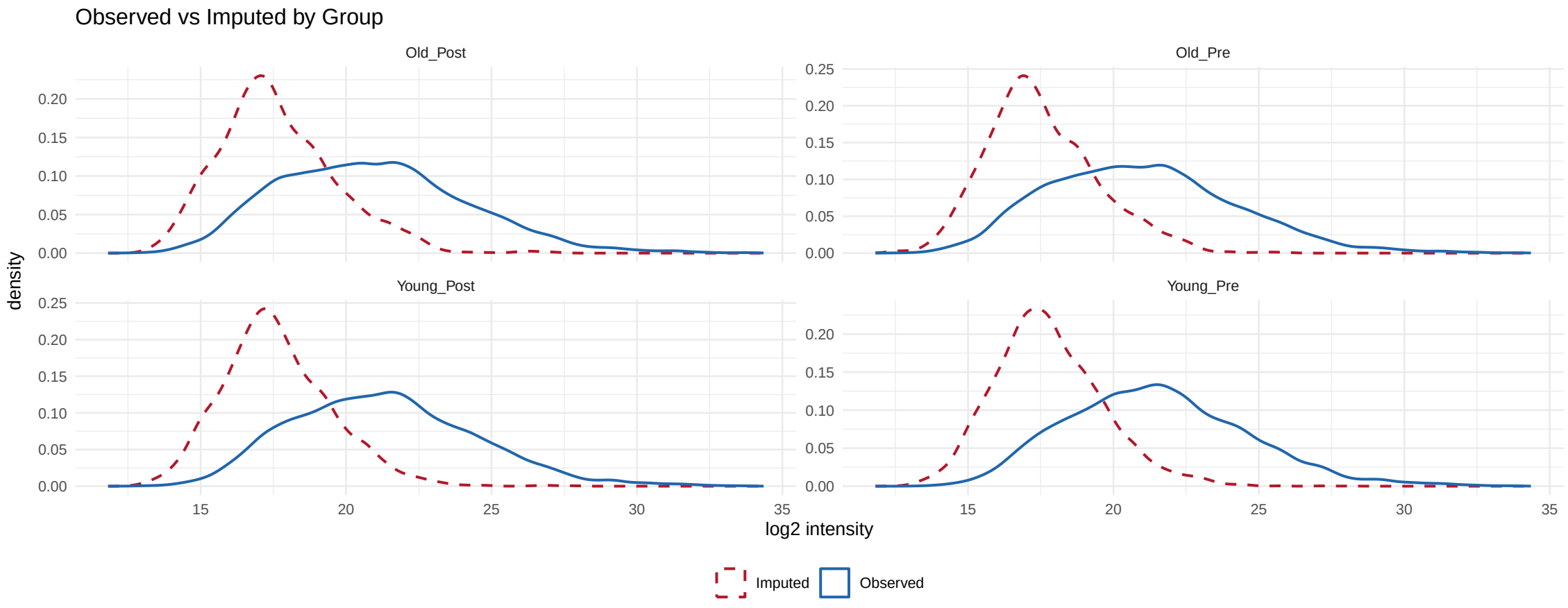
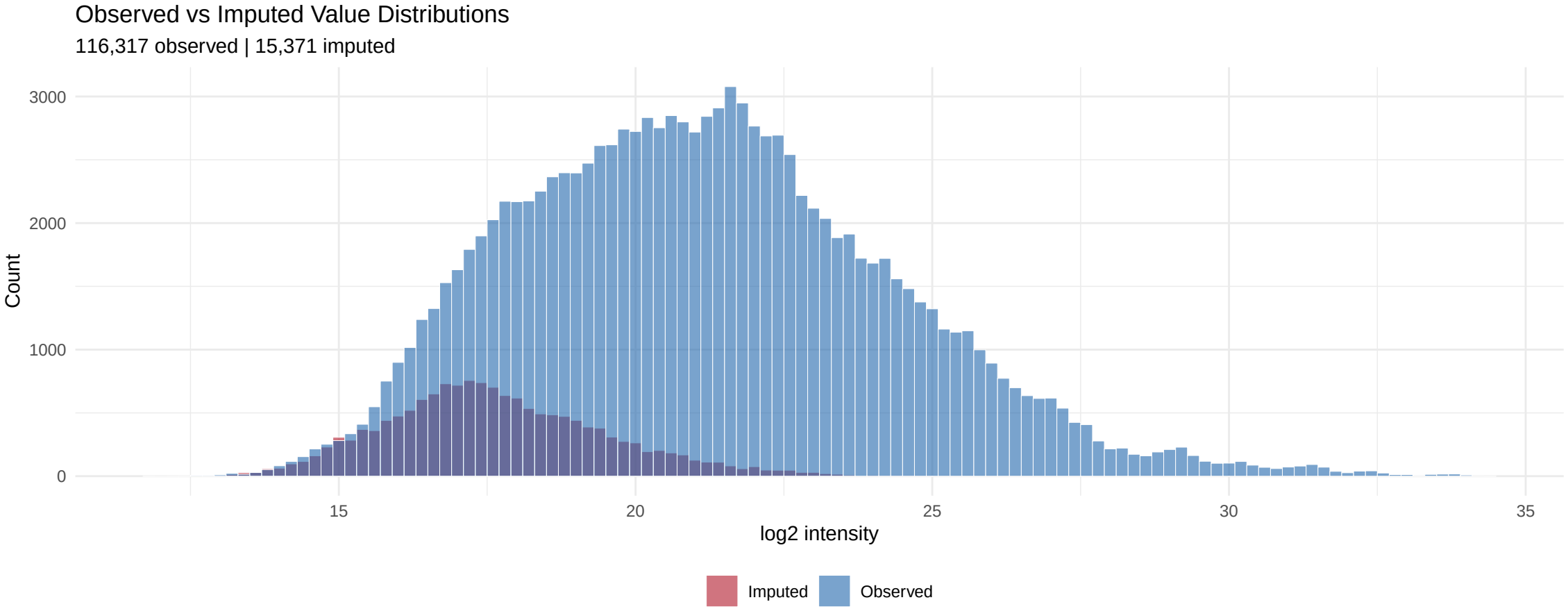
Imputation Benchmark

Imputation Benchmark
20 iter x 10% masked | Best: bpca (0.1073) | Off scale: zero (9.2)
Lower = more accurate point-wise imputation of masked MAR values

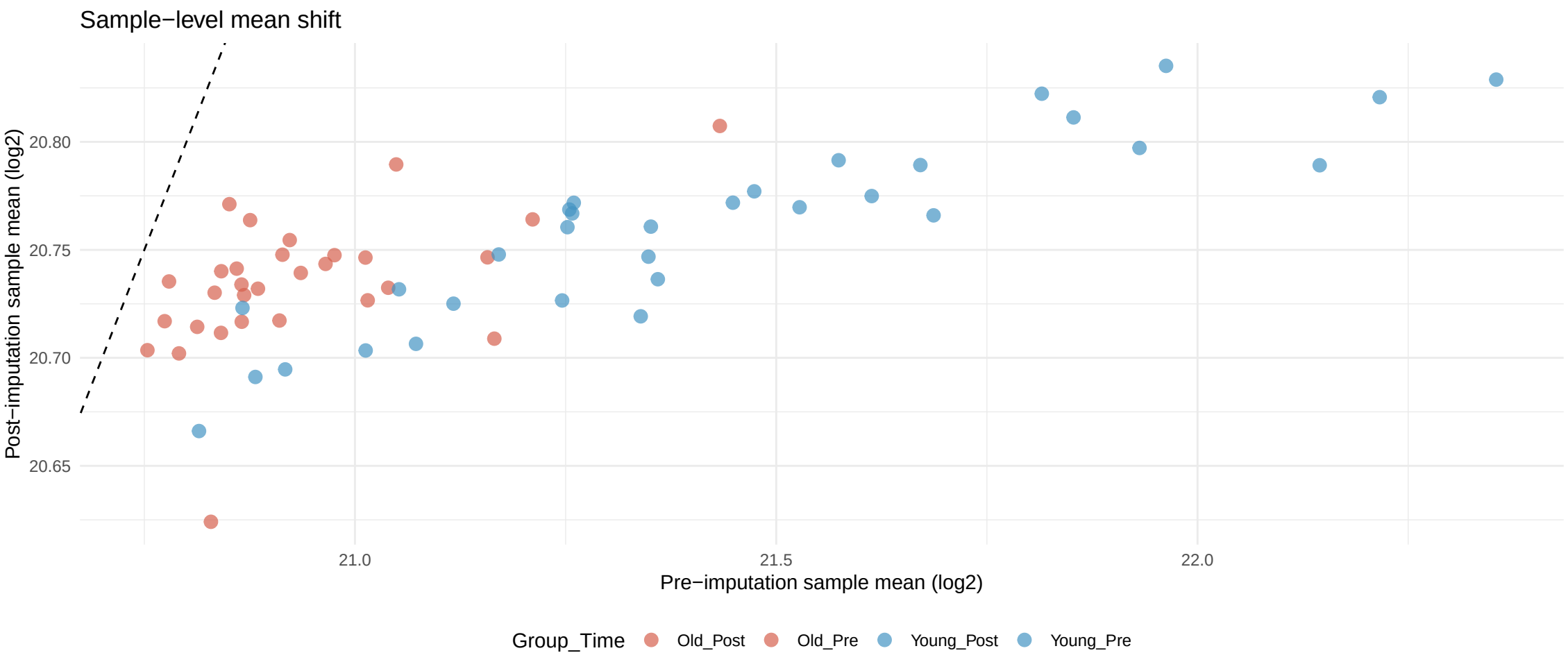
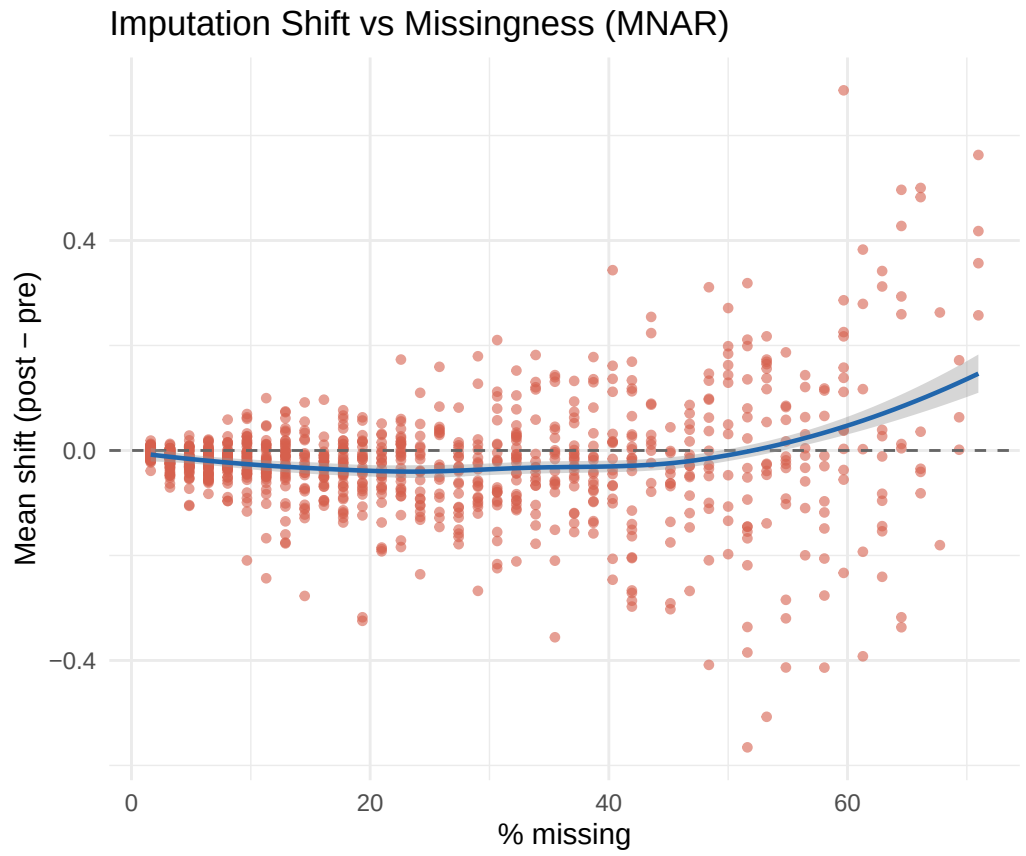


PCA Structure Preservation
Lower PSS = better preservation of global PCA geometry



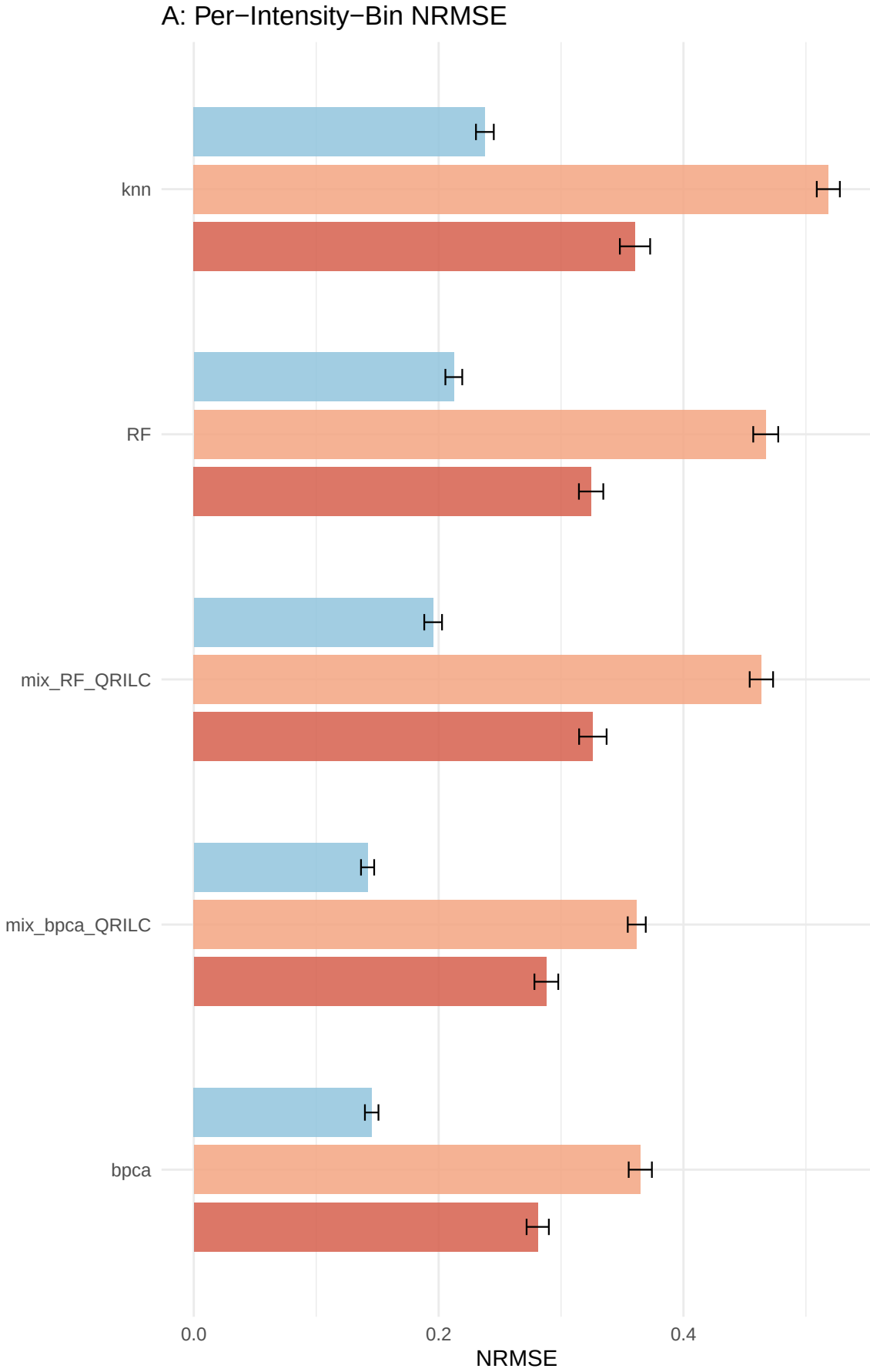


MNAR Post-Imputation Audit (1137 proteins)



Extended Diagnostics (Top 5 Methods)

Per-intensity accuracy and method summary



B: Extended Summary Ranks

Intensity tertile	method	nrmse	pss	nrmse_high	nrmse_low	nrmse_mid
	bpca	0.1073	25.7361	0.1453	0.2809	0.3647
	mix_bpca_QRILC	0.1077	23.1547	0.1420	0.2880	0.3618
	mix_RF_QRILC	0.1351	33.4078	0.1955	0.3260	0.4636
	RF	0.1398	33.4315	0.2124	0.3245	0.4672
	knn	0.1562	45.2492	0.2378	0.3604	0.5183